

Likelihood Theory

This appendix is just an overview of the likelihood theory used in this book. For greater detail or a more gentle introduction, the reader is advised to consult a book on theoretical statistics such as Cox and Hinkley (1974), Bickel and Doksum (2015), Wood (2015) or Rice (1998).

A.1 Maximum Likelihood

Consider n independent discrete random variables, $Y_1, \dots, Y_n$, with probability distribution function $f(y|\theta)$ where θ is the, possibly vector-valued, parameter. Suppose we observe $\mathbf{y} = (y_1, \dots, y_n)^T$, then we define the likelihood as:

$$P(\mathbf{Y} = \mathbf{y}) = \prod_{i=1}^n f(y_i|\theta) = L(\theta|\mathbf{y})$$

So the likelihood is a function of the parameter(s) given the data and is the probability of the observed data given a specified value of the parameter(s).

For continuous random variables, $Y_1, \dots, Y_n$ with probability density function $f(y|\theta)$, we recognize that, in practice, we can only measure or observe data with limited precision. We may record y_i , but this effectively indicates an observation in the range $[y_i^l, y_i^u]$ so that:

$$P(Y_i = y_i) = P(y_i^l \leq y_i \leq y_i^u) = \int_{y_i^l}^{y_i^u} f(u|\theta) du \approx f(y_i|\theta) \delta_i$$

where $\delta_i = y_i^u - y_i^l$. We can now write the likelihood as:

$$L(\theta|\mathbf{y}) \approx \prod_{i=1}^n f(y_i|\theta) \prod_{i=1}^n \delta_i$$

Now provided that δ_i is relatively small and does not depend on θ , we may ignore it and the likelihood is the same as in the discrete case.

As an example, suppose that Y is binomially distributed $B(n, p)$. The likelihood is:

$$L(p|\mathbf{y}) = \binom{n}{y} p^y (1-p)^{n-y}$$

The *maximum likelihood* estimate (MLE) is the value of the parameter(s) that gives the largest probability to the observed data or, in other words, maximizes the

likelihood function. The value at which the maximum occurs, $\hat{\theta}$, is the maximum likelihood estimate. In most cases, it is easier to maximize the log of likelihood function, $l(\theta|y) = \log L(\theta|y)$. Since log is a monotone increasing function, the maximum occurs at the same $\hat{\theta}$.

In a few cases, we can find an exact analytical solution for $\hat{\theta}$. For the binomial, we have the log-likelihood:

$$l(p|y) = \log \binom{n}{y} + y \log p + (n-y) \log(1-p)$$

The *score function*, $u(\theta)$, is the derivative of the log-likelihood with respect to the parameters. For this example, we have:

$$u(p) = \frac{dl(p|y)}{dp} = \frac{y}{p} - \frac{n-y}{1-p}$$

We can find the maximum likelihood estimate $\hat{p}$ by solving $u(p) = 0$. We get $\hat{p} = y/n$. We should also verify that this stationary point actually represents a maximum.

Usually we want more than an estimate; some measure of the uncertainty in the estimate is valuable. This can be obtained via the Fisher information which is:

$$I(\theta) = \text{var } u(\theta) = -E \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta^T}$$

If there is more than one parameter, $I(\theta)$ will be a matrix. The information at $\hat{\theta}$ is the second derivative at the maximum. Large values indicate high curvature so that the maximum is well defined and even close alternatives will have much lower likelihood. This would indicate a high level of confidence in the estimate. One can show that the variance of $\hat{\theta}$ can be estimated by:

$$\text{var } (\hat{\theta}) = I^{-1}(\hat{\theta})$$

under mild conditions. Sometimes it is difficult to compute the expected value of the matrix of second derivatives. As an alternative, the observed, rather than expected, value at $\hat{\theta}$ may be used instead. For the binomial example this gives:

$$\text{var } \hat{p} = \hat{p}(1-\hat{p})/n$$

We illustrate these concepts by plotting the log-likelihood for two binomial datasets: one where $n = 25, y = 10$ and another where $n = 50, y = 20$. We construct the log-likelihood function:

```
loglik <- function(p,y,n) lchoose(n,y) + y*log(p) + (n-y)*log(1-p)
```

For ease of presentation, we normalize by subtracting the log-likelihood at the maximum likelihood estimate:

```
nloglik <- function(p,y,n) loglik(p,y,n) - loglik(y/n,y,n)
```

Now plot the two log-likelihoods, as seen in Figure A.1:

```
pr <- seq(0.05,0.95,by=0.01)
matplot(pr,cbind(nloglik(pr,10,25),nloglik(pr,20,50)),type="l", xlab="
  ⇨ p",ylab="log-likelihood")
```

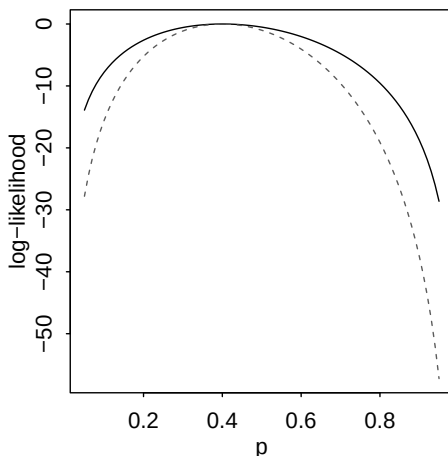


Figure A.1 Normalized binomial log-likelihood for $n = 25, y = 10$ shown with a solid line and $n = 50, y = 20$ shown with a dotted line.

We see that the maximum occurs at $p = 0.4$ in each case at a value of zero because of the normalization. For the larger sample, we see greater curvature and hence more information.

Examples where likelihood can be maximized explicitly are confined to simple cases. Typically, numerical optimization is necessary. The *Newton–Raphson* method is the most well-known technique. Let θ_0 be an initial guess at θ , then we update using:

$$\theta_1 = \theta_0 - H^{-1}(\theta_0)u(\theta_0)$$

where H is the Hessian matrix of second derivatives:

$$H(\theta) = \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta^T}$$

We iterate this method, putting θ_1 in place of θ_0 and so on, until the procedure (hopefully) converges. This method works well provided the log-likelihood is smooth and convex around the maximum and that the initial value is reasonably close. In less well-behaved cases, several things can go wrong:

- The likelihood has multiple maxima. The maximum that Newton–Raphson finds will depend on the choice of initial estimate. If you are aware that multiple maxima may exist, it is advisable to try multiple starting values to search for the overall maximum. The number and choice of these starting values are problematic. Such problems are common in fitting neural networks, but rare for generalized linear models.
- The maximum likelihood may occur at the boundary of the parameter space. This means that perhaps $u(\hat{\theta}) \neq 0$, which will confuse the Newton–Raphson method. Mixed effect models have several variance parameters. In some cases, these are maximized at zero, which causes difficulties in the numerical optimization.

- The likelihood has a large number of parameters and is quite flat in the neighborhood of the maximum. The Newton–Raphson method may take a long time to converge.

The Fisher scoring method replaces H with $-I$ and sometimes gives superior results. This method is used in fitting GLMs and is equivalent to iteratively reweighted least squares.

A minimization function that uses a Newton-type method is available in R. We demonstrate its use for likelihood maximization. Note that we need to minimize $-l$ because `nlm` minimizes, not maximizes:

```
f <- function(x) -loglik(x, 10, 25)
mm <- nlm(f, 0.5, hessian=T)
```

We use a starting value of 0.5 and find the optimum at:

```
mm$estimate
[1] 0.4
```

The inverse of the Hessian at the optimum is equal to the standard estimate of the variance:

```
c(1/mm$hessian, 0.4*(1-0.4)/25)
[1] 0.0096016 0.0096000
```

Of course, this calculation is not necessary for the binomial, but it is useful for cases where exact calculation is not possible.

A.2 Hypothesis Testing

Consider two nested models, a larger model Ω and a smaller model ω . Let $\hat{\theta}_\Omega$ be the maximum likelihood estimate under the larger model, while $\hat{\theta}_\omega$ is the corresponding value when θ is restricted to the range proscribed by the smaller model. The *likelihood ratio test statistic* (LRT) is:

$$2\log(L(\hat{\theta}_\omega)/L(\hat{\theta}_\Omega)) = -2(l(\hat{\theta}_\omega) - l(\hat{\theta}_\Omega))$$

Under some regularity conditions, this statistic is asymptotically distributed χ^2 with degrees of freedom equal to the difference in the number of identifiable parameters in the two models. The approximation may not be good for small samples and may fail entirely if the regularity conditions are broken. For example, if the smaller model places some parameters on the boundary of the parameter space, the χ^2 may not be valid. This can happen in mixed effects models when testing whether a particular variance component is zero.

The *Wald test* may be used to test hypotheses of the form $H_0 : \theta = \theta_0$ and the test statistic takes the form:

$$(\hat{\theta} - \theta_0)^T I(\hat{\theta})(\hat{\theta} - \theta_0)$$

Under the null, the test statistic has approximately a χ^2 distribution with degrees of freedom equal to the number of parameters being tested. Quite often, one does not wish to test all the parameters and the Wald test is confined to a subset. In particular, if we test only one parameter, $H_0 : \theta_i = \theta_{i0}$, the square root of the Wald test statistic is simply:

$$z = \frac{\hat{\theta}_i - \theta_{i0}}{se(\hat{\theta}_i)}$$

This is asymptotically normal. For a Gaussian linear model, these are the t -statistics and have an exact t -distribution, but for generalized linear and other models, the normal approximation must suffice.

The *score test* of the hypothesis $H_0 : \theta = \theta_0$ uses the statistic:

$$u(\theta_0)^T I^{-1}(\theta_0) u(\theta_0)$$

and is asymptotically χ^2 distributed with degrees of freedom equal to the number of parameters being tested.

There is no uniform advantage to any of these three tests. The score test does not require finding the maximum likelihood estimate, while the likelihood ratio test needs this computation to be done for both models. The Wald test needs just one maximum likelihood estimate. However, although the likelihood ratio test requires more information, the extra work is often rewarded. Although the likelihood ratio test is not always the best, it has been shown to be superior in a wide range of situations. Unless one has indications to the contrary or the computation is too burdensome, the likelihood ratio test is the recommended choice.

These test methods can be inverted to produce confidence intervals. To compute a $100(1 - \alpha)\%$ confidence interval for θ , we calculate the range of hypothesized θ_0 such that $H_0 : \theta_0 = 0$ would not be rejected at the α level. The computation is simple for the single-parameter Wald test where the confidence interval for θ_i is:

$$\hat{\theta}_i \pm z^{1-\alpha/2} se(\hat{\theta}_i)$$

where z is the appropriate quantile of the normal distribution. The computation is trickier for the likelihood ratio test. If we are interested in a confidence interval for a single parameter θ_i , we will need to compute the log-likelihood for a range of θ_i with the other θ set to the maximizing values. This is known as the *profile likelihood* for θ_i . Once this is computed as $l_i(\theta_i|y)$, the confidence interval is:

$$\{\theta_i : 2(l(\hat{\theta}_i|y) - l(\theta_i|y)) < \chi_1^{1-\alpha}\}$$

As an example, this type of calculation is used in the computation of the confidence interval for the transformation parameter used in the Box–Cox method.

We can illustrate this by considering a binomial dataset where $n = 100$ and $y = 40$. We plot the normalized log-likelihood in Figure A.2 where we have drawn a horizontal line at half the distance of the 0.95 quantile of χ_1^2 below the maximum:

```
pr <- seq(0.25, 0.55, by=0.01)
plot(pr, nloglik(pr, 40, 100), type="l", xlab="p", ylab="log-likelihood")
abline(h=-qchisq(0.95, 1)/2)
```

All p that have a likelihood above the line are contained within a 95% confidence interval for p . We can compute the range by solving for the points of intersection:

```
g <- function(x) nloglik(x, 40, 100) + qchisq(0.95, 1)/2
uniroot(g, c(0.45, 0.55))$root
[1] 0.49765
uniroot(g, c(0.25, 0.35))$root
[1] 0.30743
abline(v=c(0.49765, 0.30743))
```

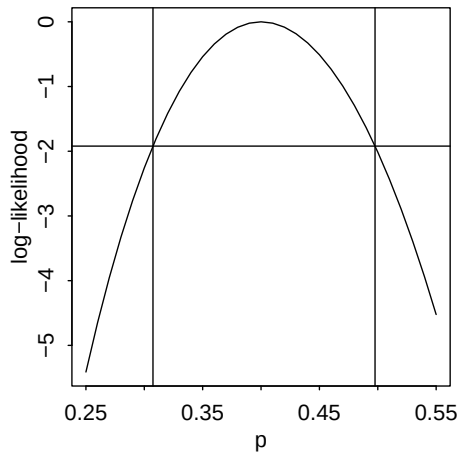


Figure A.2 Likelihood ratio test-based confidence intervals for binomial p .

The confidence interval is (0.307, 0.498) as is indicated by the vertical lines on the plot. We can compute the Wald test-based interval as:

```
se <- sqrt(0.4*(1-0.4)/100)
cv <- qnorm(0.975)
c(0.4-cv*se, 0.4+cv*se)
[1] 0.30398 0.49602
```

which is very similar, but not identical, to the LRT-based intervals.

Suppose we are interested in the hypothesis, $H_0 : p = 0.5$. The LRT and p -value are:

```
(lrstat <- 2*(loglik(0.4, 40, 100) - loglik(0.5, 40, 100)))
[1] 4.0271
pchisq(lrstat, 1, lower=F)
[1] 0.044775
```

So the null is barely rejected at the 5% level. The Wald test gives:

```
(z <- (0.5-0.4)/se)
[1] 2.0412
2*pnorm(z, lower=F)
[1] 0.041227
```

Again, not very different from the LRT. The score test takes more effort to compute. The observed information is:

$$\frac{-d^2 l(p|y)}{dp^2} = \frac{y}{p^2} + \frac{n-y}{(1-p)^2}$$

We compute the score and information at $p = 0.5$ and then form the test and get the p -value:

```
(sc <- 40/0.5 - (100-40)/(1-0.5))
[1] -40
(obsinf <- 40/0.5^2 + (100-40)/(1-0.5)^2)
[1] 400
(score.test <- 40*40/400)
[1] 4
```

```
pchisq(4, 1, lower=F)
```

```
[1] 0.0455
```

The outcome is again slightly different from the previous two tests. Asymptotically, the three tests agree. We have a moderate size sample in the example, so there is little difference. More substantial differences could be expected for smaller sample sizes.

A.3 Model Selection

It is possible to use hypothesis testing methods to choose between models but this is not always practical or desirable. Sometimes the models we want to compare are not nested and sometimes we want to compare larger numbers of models. Comparing multiple models introduces problems with multiple testing which degrades the meaning of p -values. Furthermore, we might wish to select a model for superior prediction — hypothesis testing is not designed for this purpose.

We envisage comparing several models and require a criterion that will measure the fitness and allow us to choose the best model. Many choices have been proposed but some are only applicable to particular classes of model. We present three popular criteria:

AIC or Akaike Information Criterion: The Kullback-Leibler distance is a measure of the closeness of our proposed model to the true model, having density $g(y)$. It is defined by

$$K(f, g) = \int (\log g(y) - \log(f(y|\hat{\theta})))g(y)d(y)$$

This is the expected log-likelihood ratio of the true model and the proposed model, evaluated at the true model. This would be an ideal criterion except that we need to know the true model. However, we can estimate the expected value $EK(f, g)$ using

$$-l(\hat{\theta}) + p + \int \log(g(y))g(y)dy$$

where p is the number of parameters. Since the last term only depends on the true model and not our proposed model, we can discard it when comparing proposed models. The AIC criterion is then:

$$AIC = -2l(\hat{\theta}) + 2p$$

The multiplication by two makes the criterion compatible with the definition of the likelihood ratio statistic which also multiplies by two (so that the null distribution is χ^2). AIC is not perfect — one problem is its lack of consistency. It is not guaranteed to choose the true model asymptotically. Nevertheless, it is the most widely used model selection criterion.

BIC or Bayes Information Criterion: This criterion has a Bayesian motivation. The *marginal likelihood* is defined as

$$\int f(y|\theta)h(\theta)d\theta$$

where $h(\theta)$ is a prior density on θ . The *Bayes factor* compares two models by considering the ratio of the marginal likelihoods. All things being equal, we would favor the model with the larger marginal likelihood. As we can see, this will depend on the choice of prior $h(\theta)$ which is undesirable. However, a crude approximation to minus twice the log marginal likelihood is

$$BIC = -2l(\hat{\theta}) + p \log n$$

which does not depend on the prior. We can now use BIC to compare models without concern for the prior, making it usable for non-Bayesian model selection. Notice the similarity to AIC despite the different motivation. BIC penalizes larger models more heavily than AIC and so it prefers smaller models.

DIC or Deviance Information Criterion: In more complex models, particularly involving hierarchical data structures, it is not always obvious how to count p or n . DIC uses an *effective degrees of freedom*:

$$p_D = \overline{D(\bar{\theta})} - D(\bar{\theta})$$

where D is a deviance $-2l(\theta)$. The bar in $\bar{x}$ denotes mean over x . The DIC is then defined as

$$DIC = D(\bar{\theta}) + 2p_D$$

DIC is more naturally computed for Bayesian models but can also be used for likelihood-based models where we assume a noninformative prior.